An Introduction to Artificial Intelligence

1. What is Artificial Intelligence?

Artificial Intelligence (AI) refers to the simulation of human intelligence processes by machines, especially computer systems. These processes include learning (the acquisition of information and rules for using the information), reasoning (using rules to reach approximate or definite conclusions), and self-correction.

2. Core Branches of AI

AI is a broad field that encompasses several sub-disciplines, each focusing on different capabilities:

- **Machine Learning (ML):** A subset of AI that enables systems to learn and improve from experience without being explicitly programmed.
- **Deep Learning:** A specialized form of machine learning based on artificial neural networks, designed to mimic the structural framework of the human brain.
- **Natural Language Processing (NLP):** The ability of a computer program to understand, interpret, and generate human language.
- **Computer Vision:** The field that allows computers to gain high-level understanding from digital images or videos.

3. Real-World Applications

AI technologies are rapidly transforming industries by automating routine tasks, unlocking creative potential, and extracting insights from massive datasets:

- **Healthcare:** Assisting in medical imaging diagnosis and drug discovery pipelines.
- **Finance:** Identifying fraudulent transactions and managing automated algorithmic trading.
- **Customer Service:** Powering virtual assistants and conversational chatbots to provide 24/7 support.

This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.